

A Comparative Study of Deep Semantic Segmentation Architectures for Geo-Referenced Road Extraction from Aerial Imagery

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Abstract—Accurate and up-to-date road network information is essential for navigation, urban planning, and disaster response, yet manual mapping cannot keep pace with rapidly changing landscapes. This work presents a controlled comparative study of four deep semantic segmentation architectures—U-Net, U-Net++, DeepLabV3+, and LinkNet—for automated road extraction from high-resolution aerial imagery. All models share an identical training protocol (encoder, loss, optimizer, schedule, augmentation, and data), isolating the architecture as the only experimental variable. On the Massachusetts Roads benchmark, U-Net++ achieves the best test accuracy (IoU 0.654, F_1 0.791), while LinkNet attains within 0.005 IoU of the leader using the fewest parameters and the fastest inference, revealing a practical accuracy–efficiency trade-off. Beyond pixel accuracy, we integrate a geo-referencing pipeline that converts detected road pixels into real-world coordinates using the imagery’s embedded geodetic metadata, exports vector GeoJSON, and validates the result against the OpenStreetMap road network. The best model’s geo-referenced output agrees with OpenStreetMap as closely as the human-annotated ground truth does (correctness 0.985 vs. 0.979), demonstrating that the extracted networks are directly usable in geographic information systems.

Index Terms—road extraction, semantic segmentation, remote sensing, U-Net, deep learning, geo-referencing, OpenStreetMap

I. Introduction

The continuous growth of urban environments and transportation networks has significantly increased the need for accurate, reliable, and frequently updated road information. Road networks form the backbone of modern society, supporting daily commuting, economic activity, emergency services, and large-scale infrastructure planning. As cities expand and new routes are developed, maintaining precise road data becomes both challenging and essential for effective geographic management.

Satellite and aerial imagery provide comprehensive visual coverage across vast geographic regions and have become a primary source of large-scale geographic information for infrastructure monitoring and mapping [13], [14]. Despite the availability of such rich data, extracting meaningful information from imagery remains complex: traditional manual digitization requires substantial human

effort, consumes considerable time, and cannot keep pace with rapidly changing landscapes.

Deep learning has opened new possibilities for automating the interpretation of remote sensing data. Convolutional segmentation networks recognize intricate spatial patterns, enabling fast and consistent identification of road structures at scale [1], [10]. However, published road-extraction studies typically evaluate a single architecture under author-specific training conditions, making reported accuracies difficult to compare: differences may stem from the architecture, the loss function, the augmentation policy, or simply the training budget. Moreover, most systems stop at the pixel level—a binary mask in image space—leaving a gap between detection and the geo-referenced vector data that mapping platforms and geographic information systems (GIS) actually consume.

This work addresses both gaps with two contributions:

- 1) A controlled architecture comparison. Four widely used segmentation architectures—U-Net [1], U-Net++ [2], DeepLabV3+ [3], and LinkNet [4]—are trained under a strictly identical protocol: the same ResNet-34 encoder, loss, optimizer, learning-rate schedule, augmentation, batch size, and data splits. Observed performance differences are therefore attributable to the decoder architecture alone.
- 2) An integrated geo-referencing pipeline. Detected road masks are converted to real-world coordinates using the affine transform and coordinate reference system embedded in the source GeoTIFF imagery, vectorized to GeoJSON, rendered on interactive web maps, and quantitatively validated against the OpenStreetMap (OSM) road network.

II. Related Work

Automated road extraction has been studied extensively in computer vision and remote sensing [8], [9], [13]. An early deep-learning breakthrough was DeepRoadMapper [6], which combined pixel-wise prediction from a convolutional network with graph refinement to improve topological connectivity. RoadTracer [7] instead constructed road

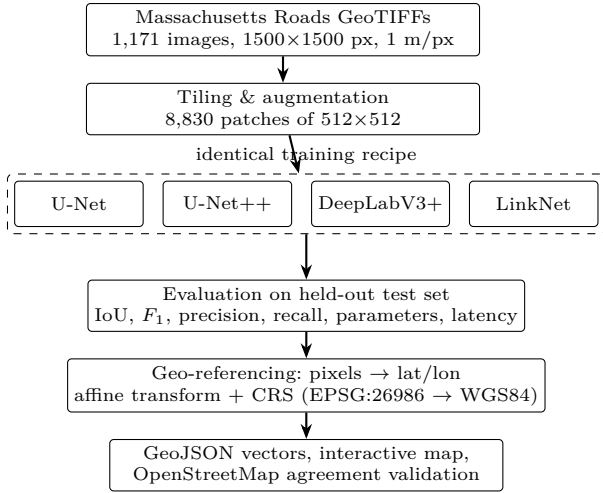


Fig. 1. System overview. Four segmentation architectures are trained under an identical protocol; the best model’s predictions are geo-referenced and validated against OpenStreetMap.

graphs by iteratively tracing paths across aerial imagery, emphasizing network formation over pixel classification.

Fully convolutional networks [10] established dense prediction as the dominant paradigm for segmentation. U-Net [1], originally designed for biomedical images, pairs a contracting encoder with a symmetric decoder and skip connections that preserve fine spatial detail; its data efficiency has made it a de facto standard for thin-structure segmentation, including roads [12]. U-Net++ [2] redesigns the skip pathways as nested, dense convolutional blocks intended to reduce the semantic gap between encoder and decoder features. DeepLabV3+ [3], [5] employs atrous (dilated) convolution and spatial pyramid pooling to aggregate multi-scale context without sacrificing resolution. LinkNet [4] targets efficiency, linking each encoder stage directly to its decoder counterpart to recover spatial information with few parameters.

Research gap. These studies demonstrate strong detection accuracy, but cross-paper comparisons are confounded by differing training conditions, and limited attention has been paid to associating detections with precise geographic coordinates. Outputs confined to the image domain are not directly usable in mapping platforms. The present work contributes a like-for-like architecture benchmark and closes the loop from pixels to geo-referenced, OSM-validated vector data.

III. Methodology

Fig. 1 summarizes the proposed framework, from raw georeferenced imagery to validated vector road networks.

A. Dataset and Preprocessing

Experiments use the Massachusetts Roads Dataset [9], comprising 1,171 aerial images of 1500×1500 pixels at 1 m/pixel resolution with pixel-level binary road annotations, split into 1,108 training, 14 validation, and 49 test

images. The imagery is distributed as GeoTIFF with an embedded affine transform in the NAD83 Massachusetts State Plane coordinate system (EPSG:26986), which our geo-referencing stage exploits directly.

Each image is cut into a 3×3 grid of 512×512 patches with slight overlap. Patches in which more than 25% of pixels are blank (white no-data padding present at collection boundaries) are discarded, yielding 8,263 training, 126 validation, and 441 test tiles. Training-time augmentation comprises horizontal/vertical flips, 90° rotations, and random brightness-contrast jitter; images are normalized with ImageNet statistics.

B. Architectures Compared

All four models are encoder-decoder segmentation networks instantiated with the same ImageNet-pretrained ResNet-34 encoder [11], differing only in decoder design: U-Net’s symmetric decoder with plain skip connections [1]; U-Net++’s nested, densely connected skip pathways [2]; DeepLabV3+’s atrous spatial pyramid pooling [3]; and LinkNet’s lightweight additive decoder [4].

C. Identical Training Protocol

Every model is trained with the combined Dice and binary cross-entropy objective

$$\mathcal{L} = \frac{1}{2} \mathcal{L}_{\text{Dice}} + \frac{1}{2} \mathcal{L}_{\text{BCE}}, \quad (1)$$

which is well suited to thin structures occupying a small fraction of pixels (roads cover $\sim 5\%$ of the dataset). Optimization uses AdamW (learning rate 3×10^{-4} , weight decay 10^{-4}) with cosine annealing over 30 epochs at batch size 8 and a fixed random seed. The checkpoint with the best validation IoU is retained. Training was performed on freely available NVIDIA T4 GPUs (Kaggle), each model completing in 2.5–9 hours.

D. Evaluation Metrics

Let TP, FP, and FN denote pixel-level true positives, false positives, and false negatives aggregated over the test set. We report intersection over union $\text{IoU} = \text{TP} / (\text{TP} + \text{FP} + \text{FN})$, F_1 (Dice), precision, and recall, together with parameter counts and per-tile inference latency.

E. Geo-Referencing and OSM Validation

The predicted binary mask of a full image is vectorized into polygons, whose vertices are mapped from pixel space to projected coordinates by the GeoTIFF affine transform and then to WGS84 longitude/latitude. The result is exported as GeoJSON and rendered over OpenStreetMap tiles for interactive inspection.

For quantitative validation, all OSM ways tagged highway within the image footprint are retrieved via the Overpass API, projected to the image grid, and buffered by 6 m. We then compute correctness, the fraction of predicted road pixels lying within the buffered OSM network (does the model invent roads?), and completeness, the fraction of the buffered OSM network covered by the prediction

TABLE I

Test-Set Comparison on the Massachusetts Roads Dataset (441 tiles; identical training protocol). Latency measured per 512×512 tile on an Apple-silicon MPS device.

Model	IoU	F_1	Prec.	Rec.	Par. (M)	ms
U-Net++	0.654	0.791	0.808	0.774	26.1	199.8
U-Net	0.650	0.788	0.807	0.770	24.4	110.5
LinkNet	0.649	0.787	0.803	0.771	21.8	105.0
DeepLabV3+	0.642	0.782	0.798	0.766	22.4	117.5

(how much of the mapped network is recovered?). Because OSM’s highway tag includes footpaths and service ways that are frequently invisible in 1m aerial imagery, we additionally report the same measures for the human-annotated ground truth as an achievable upper reference.

IV. Results and Discussion

A. Segmentation Accuracy

Table I reports test-set performance. All four architectures land in a narrow, competitive band (IoU 0.642–0.654), consistent with published results on this benchmark and confirming that, under equal training conditions, decoder design produces measurable but moderate differences.

U-Net++ attains the best accuracy on every metric, supporting the hypothesis that its nested skip pathways reduce the encoder–decoder semantic gap and benefit thin-structure localization—but at nearly double the inference latency of the alternatives. LinkNet is the clear efficiency choice: within 0.005 IoU of the leader while using 16% fewer parameters and running $1.9\times$ faster than U-Net++. DeepLabV3+, whose atrous pyramid excels on large-object benchmarks, trails here; aggressive context aggregation appears less advantageous for one-meter-wide linear structures than high-resolution skip connections. Precision exceeds recall for all models, indicating conservative predictors whose residual errors are dominated by missed narrow or occluded roads rather than hallucinated ones.

Fig. 2 shows validation IoU over training. Convergence is smooth for all models; U-Net leads early, but U-Net++ continues improving and overtakes it after epoch ~ 25 , suggesting the denser decoder benefits from a longer optimization horizon.

Fig. 3 presents a qualitative comparison. All models recover the interchange and arterial roads; disagreements concentrate in tree-occluded residential streets—the same regions where the models’ recall deficit originates.

B. Geo-Referenced Output and OSM Agreement

Applying the geo-referencing stage to the U-Net++ prediction of the Fig. 3 scene (42.321° – 42.335° N, 71.239° – 71.258° W) yields 28 road polygons in GeoJSON, rendered interactively over OpenStreetMap. Table II summarizes the agreement between the geo-referenced prediction and 916 OSM ways retrieved for the scene.

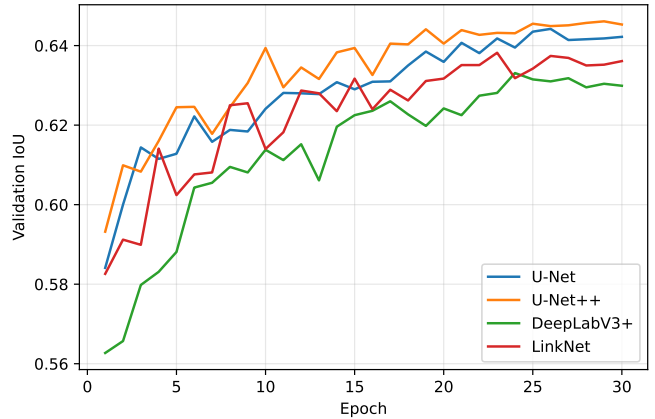


Fig. 2. Validation IoU per epoch under the identical training protocol. All models converge stably; U-Net++ overtakes U-Net in the final third of training.

TABLE II

Agreement with the OpenStreetMap Road Network (6 m buffer).

Source	Correctness	Completeness
U-Net++ prediction	0.985	0.320
Human ground truth	0.979	0.316

Two observations follow. First, correctness of 0.985 means virtually every pixel the model labels as road lies within 6 m of a mapped OSM road: the geo-referencing chain (prediction $\rightarrow$ affine transform $\rightarrow$ datum conversion) is spatially accurate, and the model does not hallucinate geometry. Second, the prediction’s completeness (0.320) matches the human ground truth’s (0.316): the apparently low value is explained by OSM footpaths, driveways, and service ways that are absent from both the imagery-derived annotation and the prediction, not by model failure. By this measure, the automatically extracted, geo-referenced network is as consistent with OSM as the manual annotation it was trained on.

V. Conclusion

This paper presented a controlled comparison of four semantic segmentation architectures for road extraction, trained under a strictly identical protocol on the Massachusetts Roads Dataset, together with a geo-referencing pipeline that converts predictions into GIS-ready, OSM-validated vector data. U-Net++ achieved the best accuracy (test IoU 0.654), while LinkNet offered near-identical accuracy at the lowest computational cost, a relevant result for large-area or on-board processing. The geo-referenced output of the best model agreed with OpenStreetMap as closely as the human ground truth, demonstrating end-to-end fidelity from pixels to coordinates.

Future work includes topology-aware losses and relaxed (buffer-based) evaluation to better reflect connectivity,

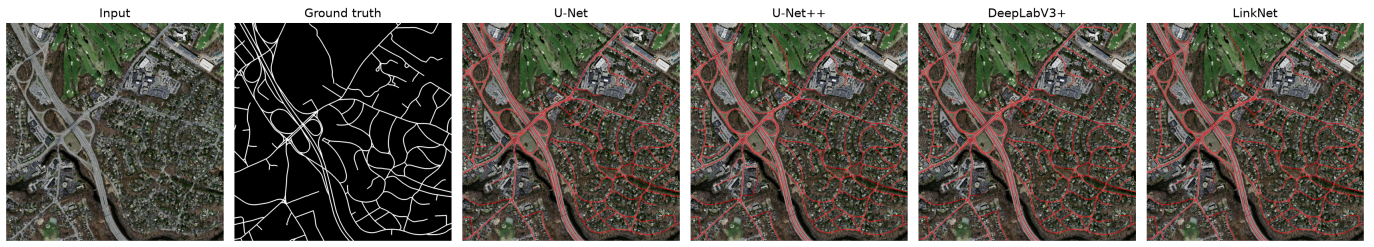


Fig. 3. Qualitative comparison on a full 1500×1500 test image (suburban Newton, MA, including a highway interchange). Left to right: input image, human ground truth, and predictions (red overlay) of the four models. All architectures recover the arterial network; differences concentrate in narrow residential streets and partially occluded segments.

transformer-based encoders, skeletonization to centerline vectors for direct map conflation, and scaling the pipeline to national road inventories using satellite constellations.

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